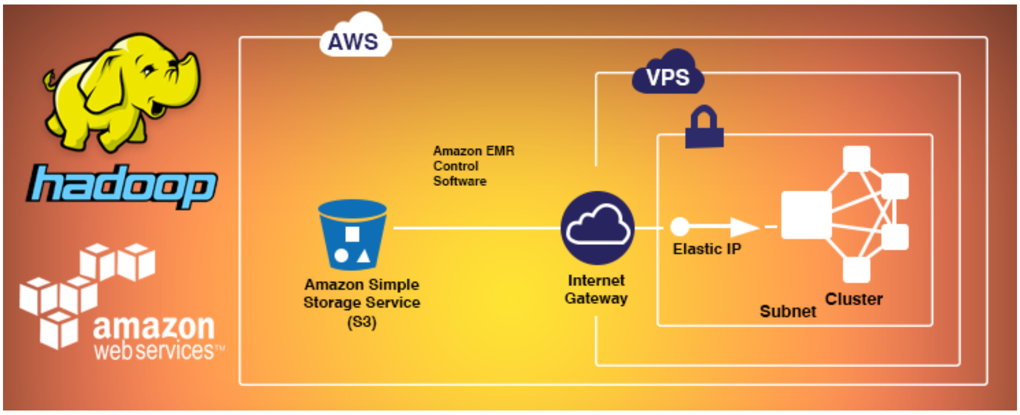
Abhishek (https://www.eduonix.com/blog/author/rony/)

July 9, 2015, 9 months ago

2 (https://www.eduonix.com/blog/bigdata-and-hadoop/a-step-by-step-guide-to-install-hadoop-cluster-on-amazon-ec2/#comments)

0 4.9K

(http://www.eduonix.com/blog/wp-content/uploads/2015/07/Setting-Hadoop-Cluster-on-Amazon-EC2-Instance.png)



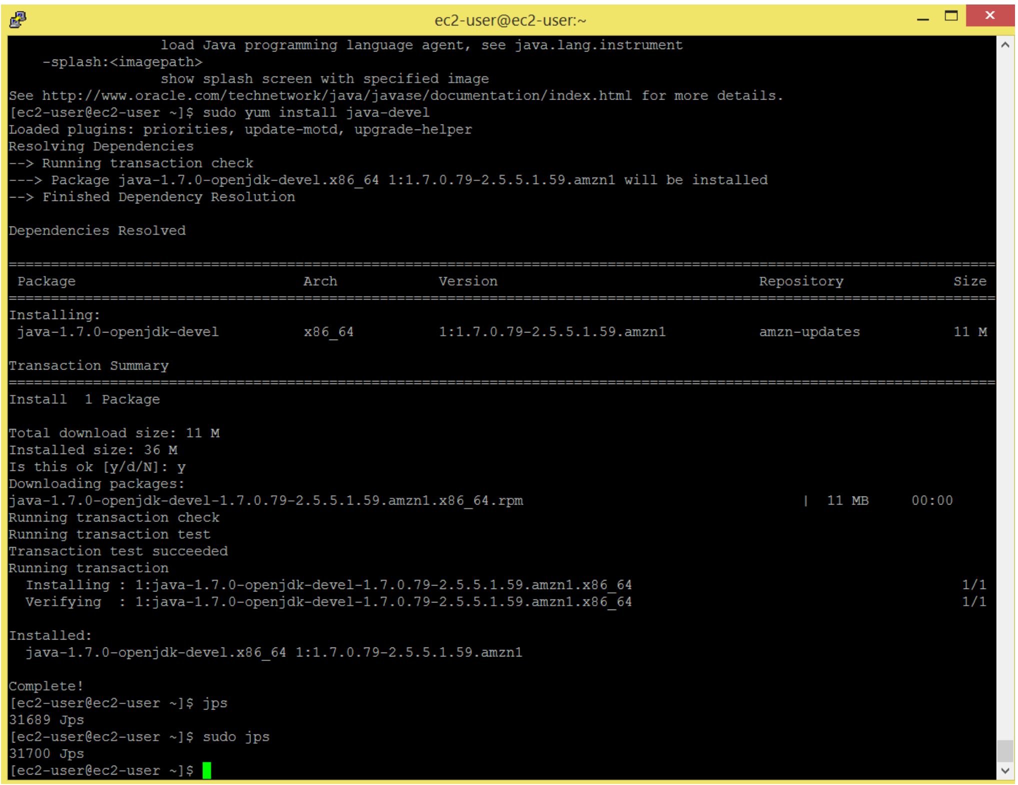
(http://www.eduonix.com/blog/wp-content/uploads/2015/07/Setting-Hadoop-Cluster-on-Amazon-EC2-Instance.png)

This is a step by step guide to install a Hadoop cluster on Amazon EC2. I have my AWS EC2 instance **ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com** ready on which I will install and configure Hadoop, java 1.7 is already installed.

In case java is not installed on you AWS EC2 instance, use below commands:

Command: sudo yum install java-1.7.0-openjdk

Command: sudo yum install java-devel
(http://www.eduonix.com/blog/wp-content/uploads/2015/07/110.png)

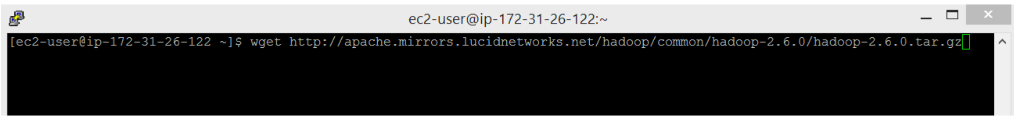


(http://www.eduonix.com/blog/wp-content/uploads/2015/07/110.png)

I am installing hadoop-2.6.0 on the cluster. Below command will download hadoop-2.6.0 package.

Command: wget
http://apache.mirrors.lucidnetworks.net/hadoop/common/had

2.6.0/hadoop-2.6.0.tar.gz

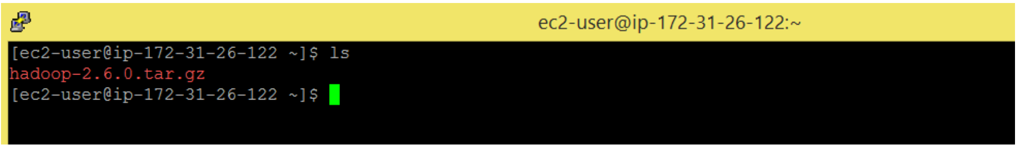


[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/25.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/25.png)

[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/34.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/34.png)

Check if the package got downloaded.

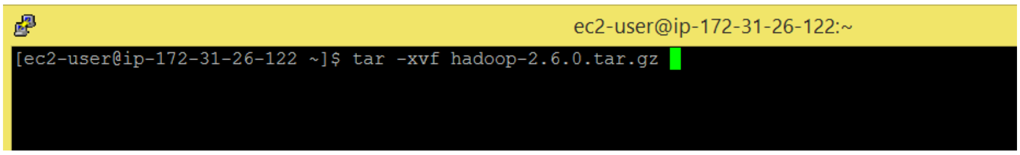
Command: ls



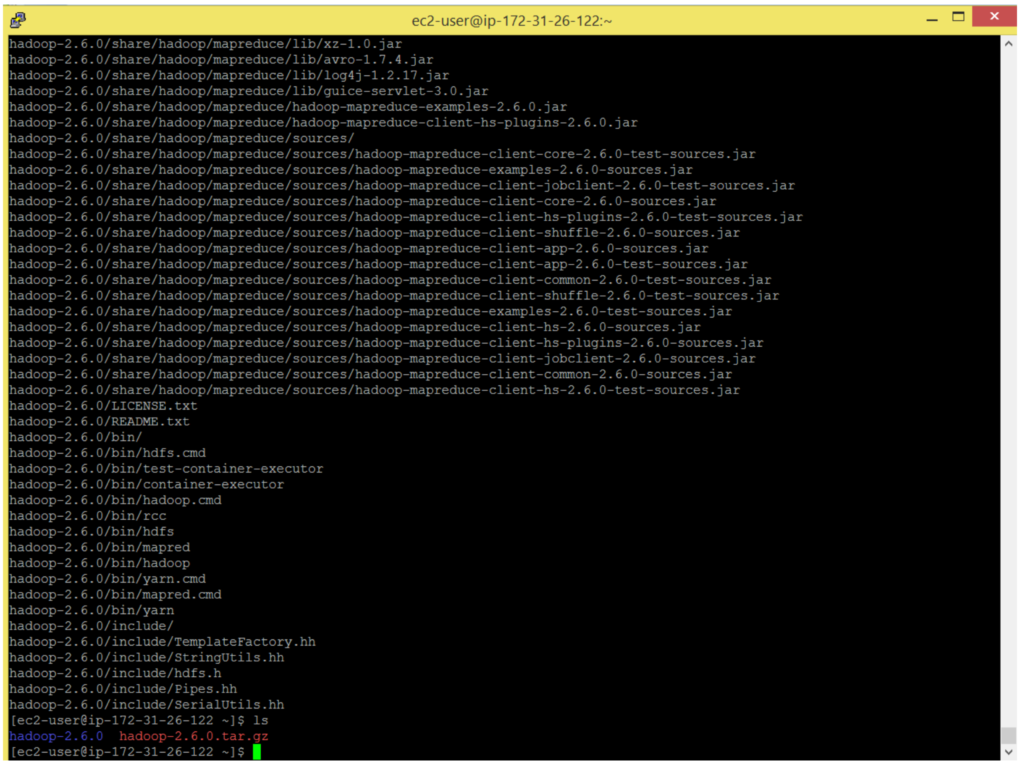
[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/43.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/43.png)

Untar the file.

Command: tar -xvf hadoop-2.6.0.tar.gz



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/52.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/52.png)

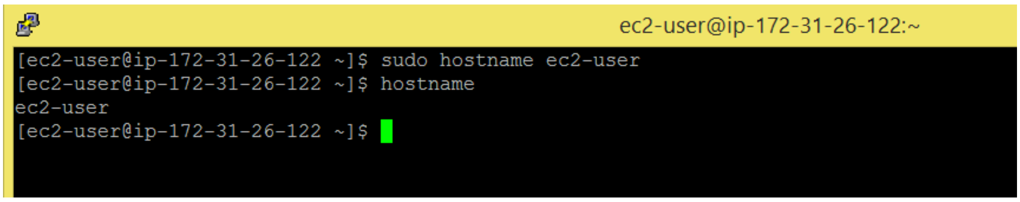


[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/63.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/63.png)

Make hostname as ec2-user

Command: sudo hostname ec2-user

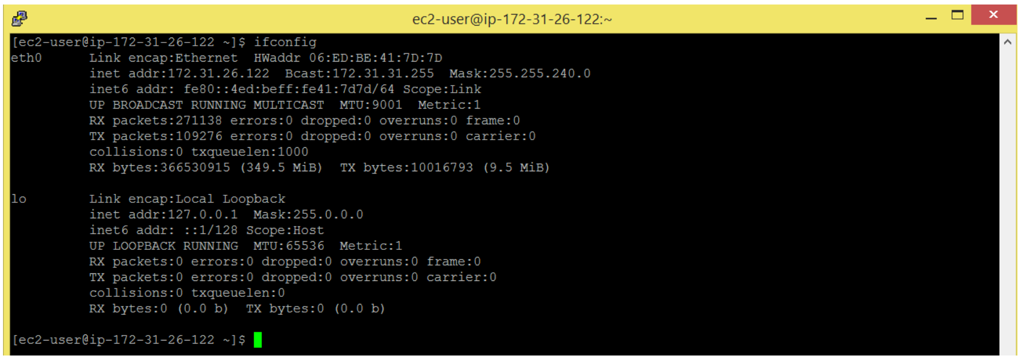
Command: hostname



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/73.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/73.png)

Below command will give you ip address, for me its 172.31.26.122

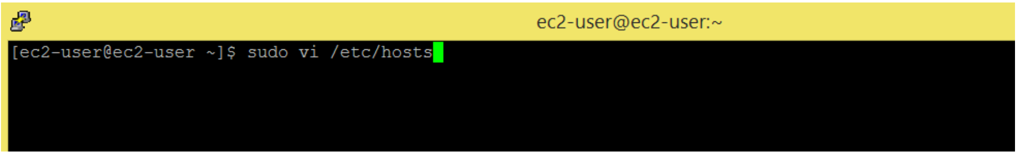
Command: ifconfig



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/83.png)

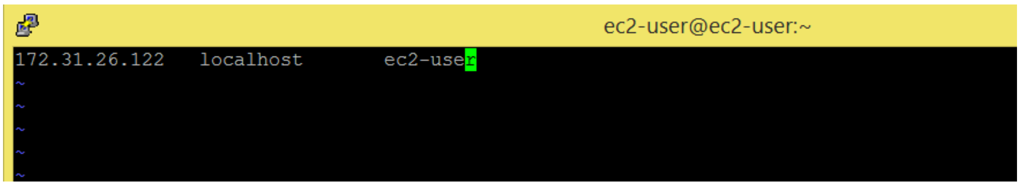
Edit /etc/hosts file.

Command: sudo vi /etc/hosts



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/93.png)

Put ip address and hostname as below, save the file and close it.



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/103.png)

The **'ssh-agent'** is a background program that handles passwords for SSH private keys.

The **'ssh-add'** command prompts the user for a private key password and adds it to the list maintained by ssh-agent. Once you add a password to ssh-agent, you will not be asked to provide the key when using SSH or SCP to connect to hosts with your public key.

You will get **.pem** file from your amazon instance settings, copy it on amazon cluster, I have copied **hadoop.pem** file on my amazon ec2 cluster.

Protect key files to avoid any accidental or intentional corruption.

Command: chmod 644 .ssh/authorized_keys
Command: chmod 400 hadoop.pem

Start ssh-agent

Note: Make sure you use the backquote
`(  ), located under the tilde ( ~ ), rather than the si`

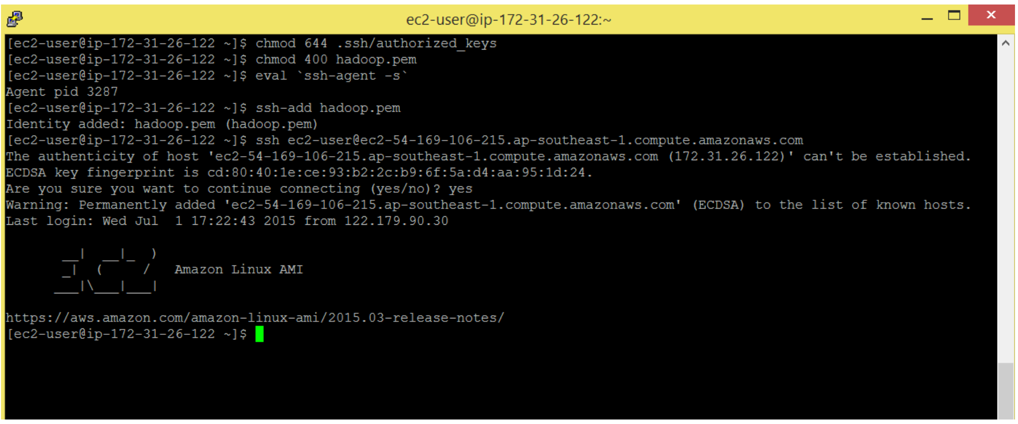
Command: `eval` ssh-agent -s `

Add the secure identity to SSH Agent Key repository

Command: ssh-add hadoop.pem

Command: ssh ec2-user@ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com

You will be able to login without password.

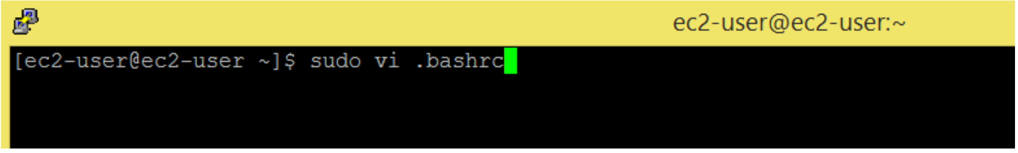


(http://www.eduonix.com/blog/wp-content/uploads/2015/07/113.png)
Come out of the login.

Command: exit

Now we will add hadoop and java environment variables in **.bashrc** file.

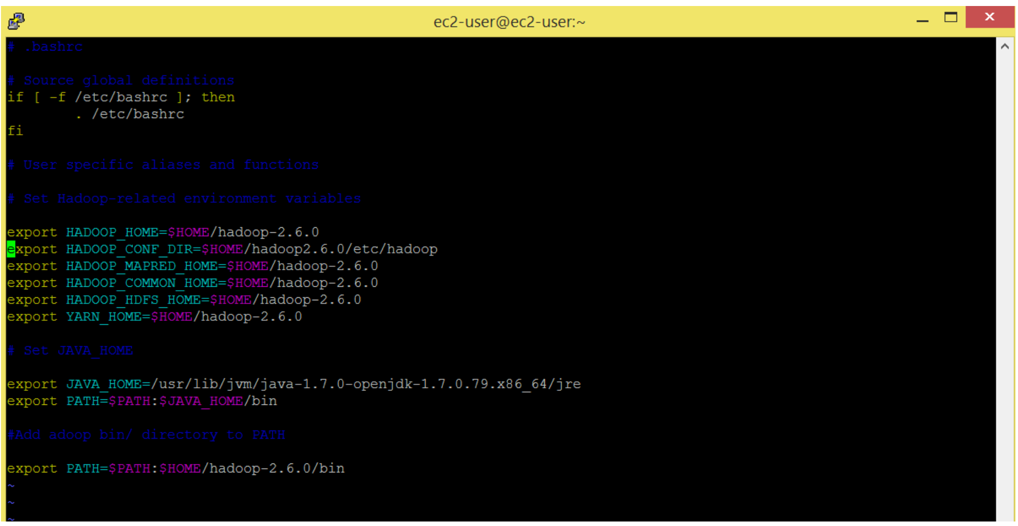
Command: sudo vi .bashrc



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/123.png)

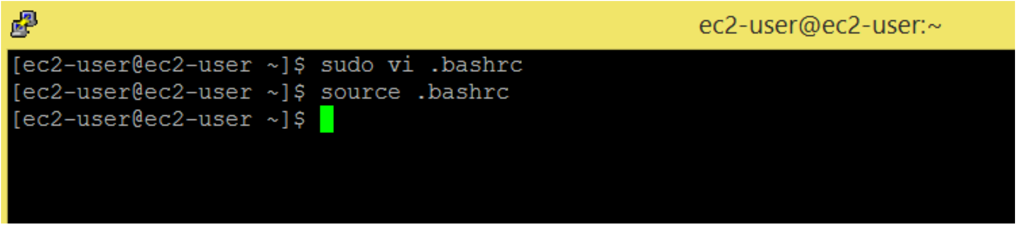
```
# Set Hadoop-related environment variables

export HADOOP_HOME=$HOME/hadoop-2.6.0

export HADOOP_CONF_DIR=$HOME/hadoop2.6.0/etc/hadoop

export HADOOP_MAPRED_HOME=$HOME/hadoop-2.6.0

export HADOOP_COMMON_HOME=$HOME/hadoop-2.6.0

export HADOOP_HDFS_HOME=$HOME/hadoop-2.6.0

export YARN_HOME=$HOME/hadoop-2.6.0

 

# Set JAVA_HOME

export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_

export PATH=$PATH:$JAVA_HOME/bin

 

# Add Hadoop bin/ directory to PATH

export PATH=$PATH:$HOME/hadoop-2.6.0/bin
```

(http://www.eduonix.com/blog/wp-content/uploads/2015/07/132.png)



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/132.png)

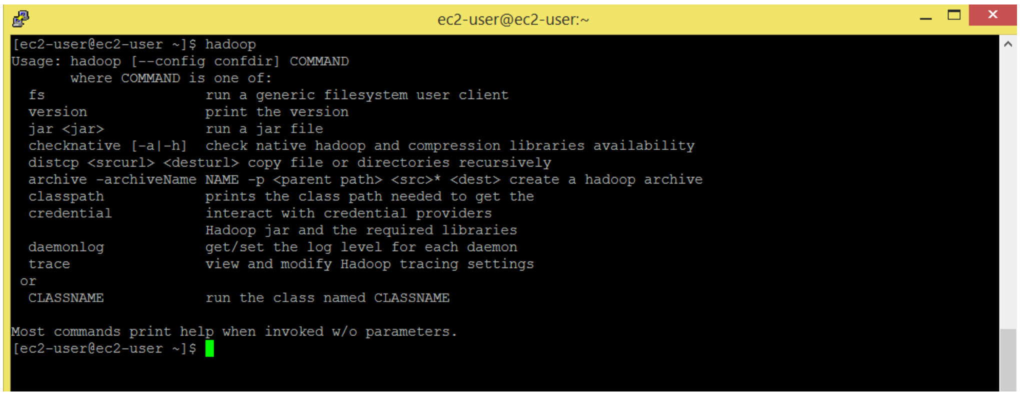
Command: source .bashrc



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/142.png)

Enter hadoop command and check if you get a set of options.

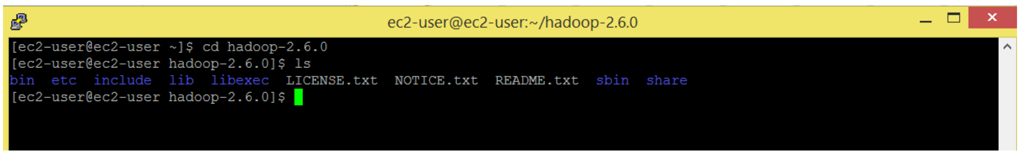
Command: hadoop



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/151.png)

check out the directories present in hadoop-2.6.0

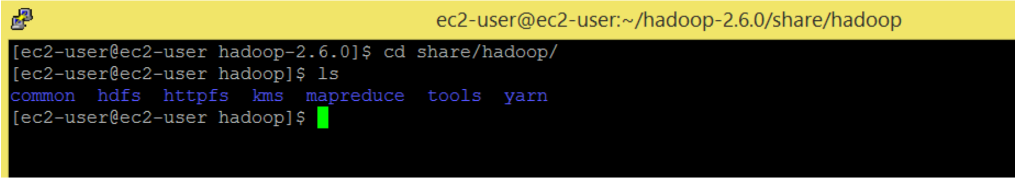
Command: cd hadoop-2.6.0



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/161.png)

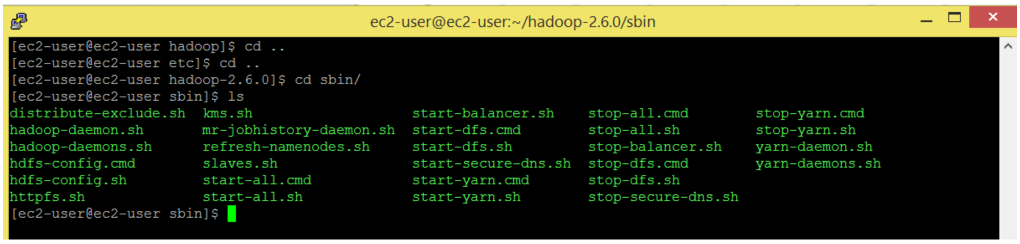
share directory contains all the jar files.

Command: cd share/hadoop



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/171.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/171.png)

sbin directory contains all the script files to run or stop hadoop daemons/cluster.



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/181.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/181.png)

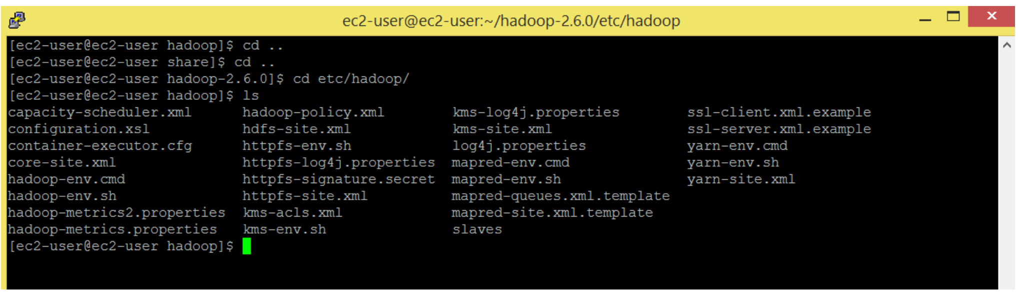
etc directory contains all the configuration files. We will edit few configuration files. Go to etc/hadoop/ directory.

Command: cd ..

Command: cd ..

Command: cd etc/hadoop

Command: ls

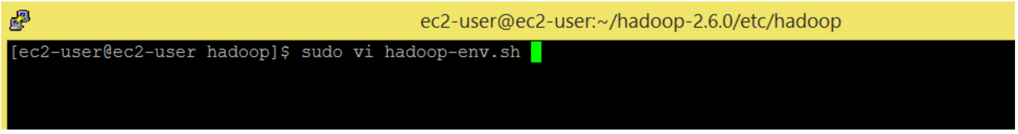


[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/191.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/191.png)

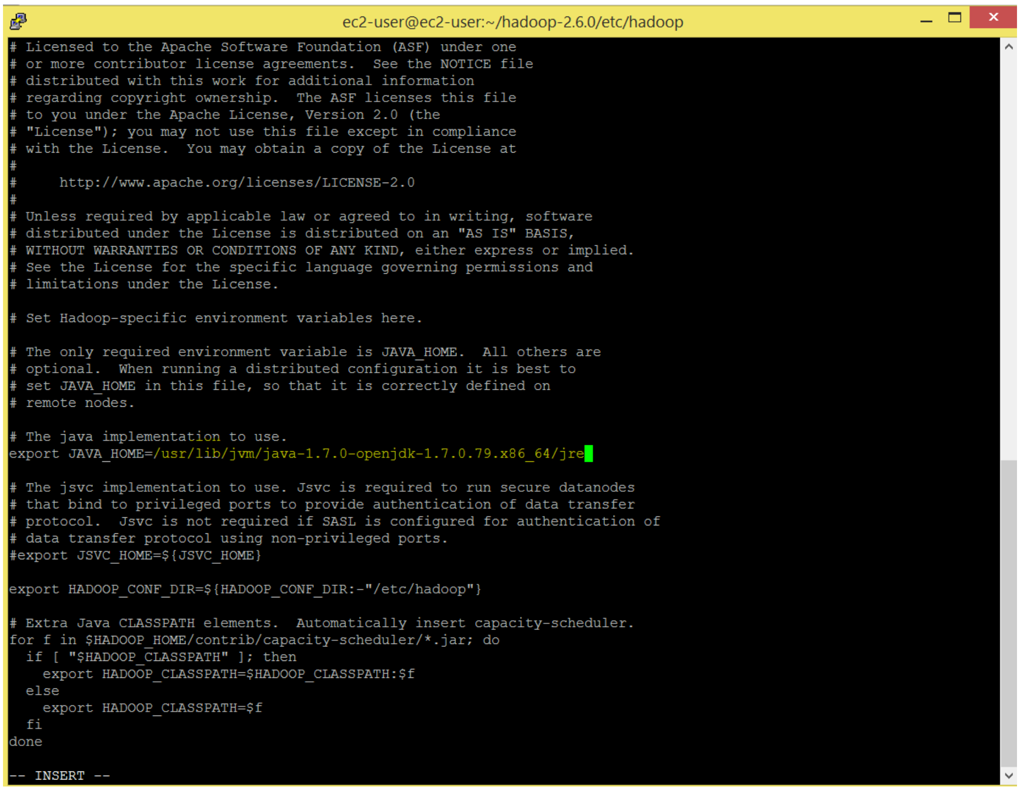
Set java home in **hadoop-env.sh** file.

```
# Set JAVA_HOME

export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_6
```

Command: sudo vi hadoop-env.sh
(http://www.eduonix.com/blog/wp-content/uploads/2015/07/201.png)



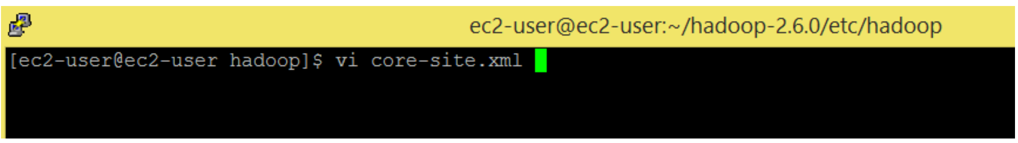
(http://www.eduonix.com/blog/wp-content/uploads/2015/07/201.png)


ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop

```
# Licensed to the Apache Software Foundation (ASF) under one
# or more contributor license agreements.  See the NOTICE file
# distributed with this work for additional information
# regarding copyright ownership.  The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License.  You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# Set Hadoop-specific environment variables here.

# The only required environment variable is JAVA_HOME.  All others are
# optional.  When running a distributed configuration it is best to
# set JAVA_HOME in this file, so that it is correctly defined on
# remote nodes.

# The java implementation to use.
export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_64/jre

# The jsvc implementation to use. Jsvc is required to run secure datanodes
# that bind to privileged ports to provide authentication of data transfer
# protocol.  Jsvc is not required if SASL is configured for authentication of
# data transfer protocol using non-privileged ports.
#export JSVC_HOME=${JSVC_HOME}

export HADOOP_CONF_DIR=${HADOOP_CONF_DIR:-"/etc/hadoop"}

# Extra Java CLASSPATH elements.  Automatically insert capacity-scheduler.
for f in $HADOOP_HOME/contrib/capacity-scheduler/*.jar; do
  if [ "$HADOOP_CLASSPATH" ]; then
    export HADOOP_CLASSPATH=$HADOOP_CLASSPATH:$f
  else
    export HADOOP_CLASSPATH=$f
  fi
done

-- INSERT --
```

(http://www.eduonix.com/blog/wp-content/uploads/2015/07/212.png)
Edit core-site.xml file. This file contains the configuration settings for Hadoop Core such as I/O settings that are common to HDFS and MapReduce.

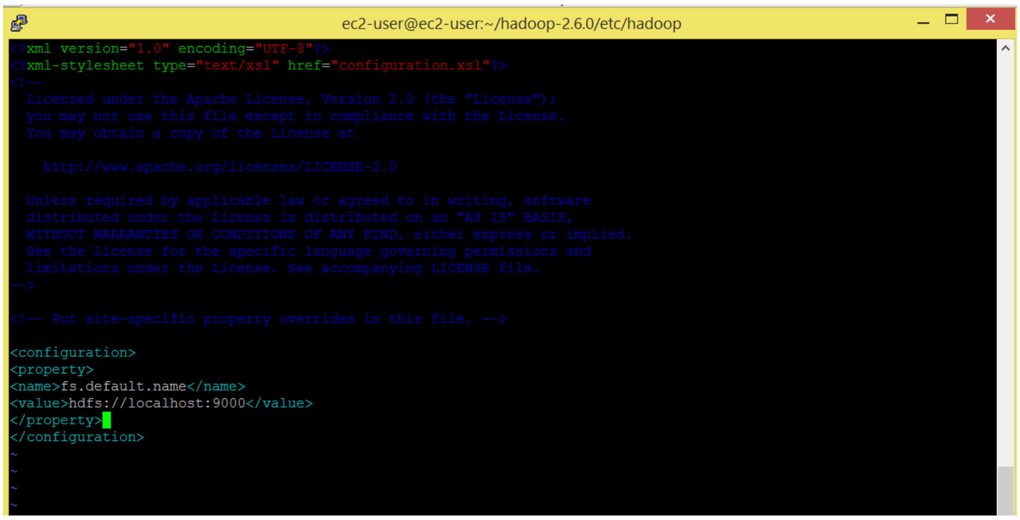
Command: vi core-site.xml



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/221.png)

```
<strong>&lt;configuration&gt; </strong>

<strong>&lt;property&gt; </strong>

<strong>&lt;name&gt;fs.default.name&lt;/name&gt; </strong>

<strong>&lt;value&gt;hdfs://localhost:9000&lt;/value&gt; </str

<strong>&lt;/property&gt; </strong>

<strong>&lt;/configuration&gt;</strong>
```

(http://www.eduonix.com/blog/wp-content/uploads/2015/07/231.png)

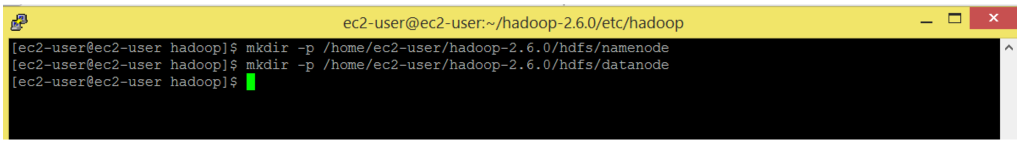


(http://www.eduonix.com/blog/wp-content/uploads/2015/07/231.png)

Make namenode and datanode directory.

Command: mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/namenode

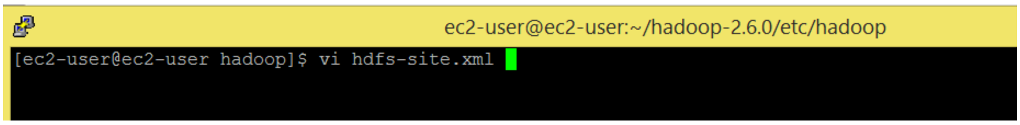
Command: mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/datanode



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/241.png)

Edit hdfs-site.xml. This file contains the cconfiguration settings for HDFS daemons; the Name Node, the secondary Name Node, and the data node.

Command: vi hdfs-site.xml



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/251.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/251.png)

```
<configuration>

<property>

<name>dfs.replication</name>

<value>1</value>

</property>

<property>

<name>dfs.namenode.name.dir</name>

<value>/home/ec2-user/hadoop-2.6.0/hdfs/namenode</value>

</property>

<property>

<name>dfs.datanode.data.dir</name>

<value>>/home/ec2-user/hadoop-2.6.0/hdfs/datanode</value>

</property>

</configuration>
```

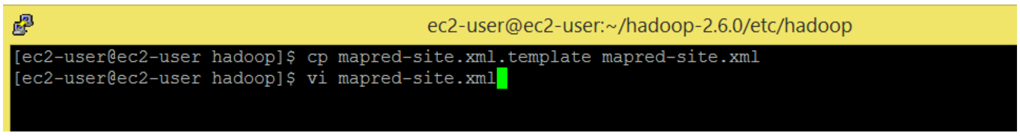
[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/26.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/26.png)

(http://www.eduonix.com/blog/wp-content/uploads/2015/07/26.png)

Edit mapred-site.xml. This file contains the configuration settings for MapReduce daemons.

Command: cp mapred-site.xml.template mapred-site.xml

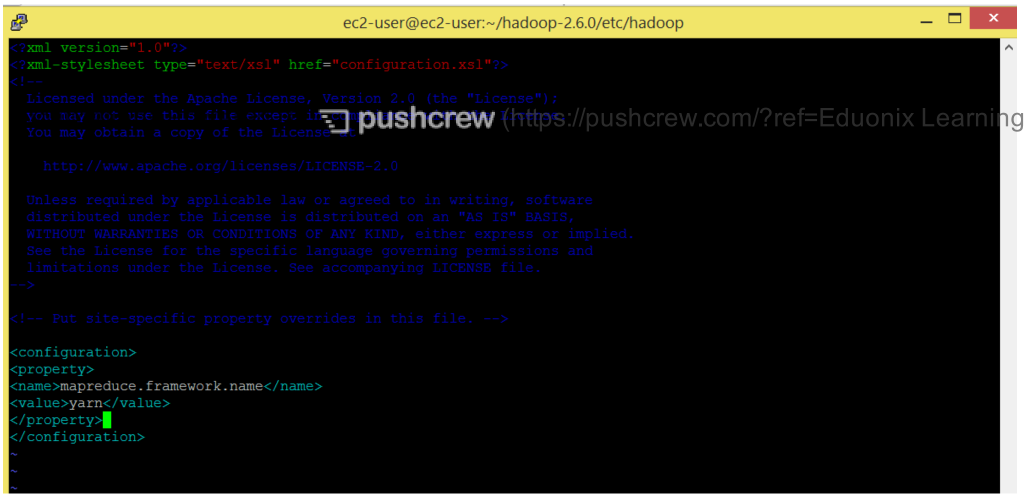
Command: vi mapred-site.xml



(http://www.eduonix.com/blog/wp-content/uploads/2015/07/27.png)

```
<configuration>

<property>

<name>mapreduce.framework.name</name>

<value>yarn</value>

</property>

</configuration>
```

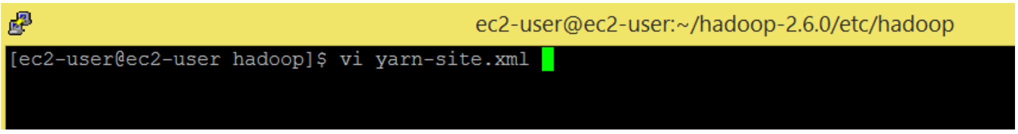
(http://www.eduonix.com/blog/wp-content/uploads/2015/07/28.png)



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/28.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/28.png)

Edit yarn-site.xml. This file contains the configuration settings for YARN.

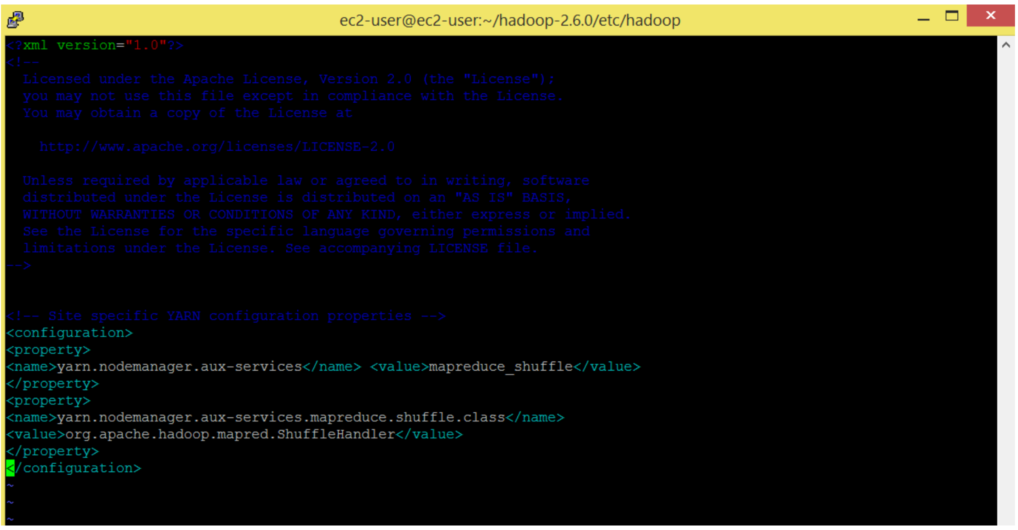
Command: vi yarn-site.xml



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/29.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/29.png)

```
<configuration>
<property>
 <name>yarn.nodemanager.aux-services</name> <value>mapreduce_sh
</property>
<property>
<name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</na
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
```

[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png)

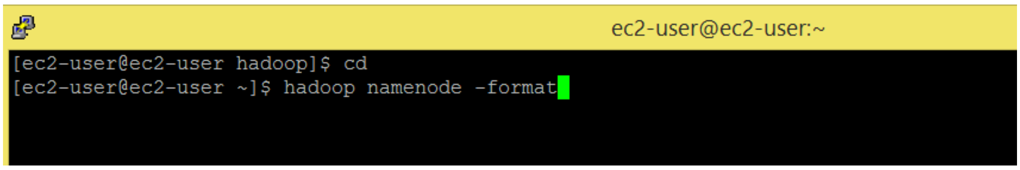


[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png)

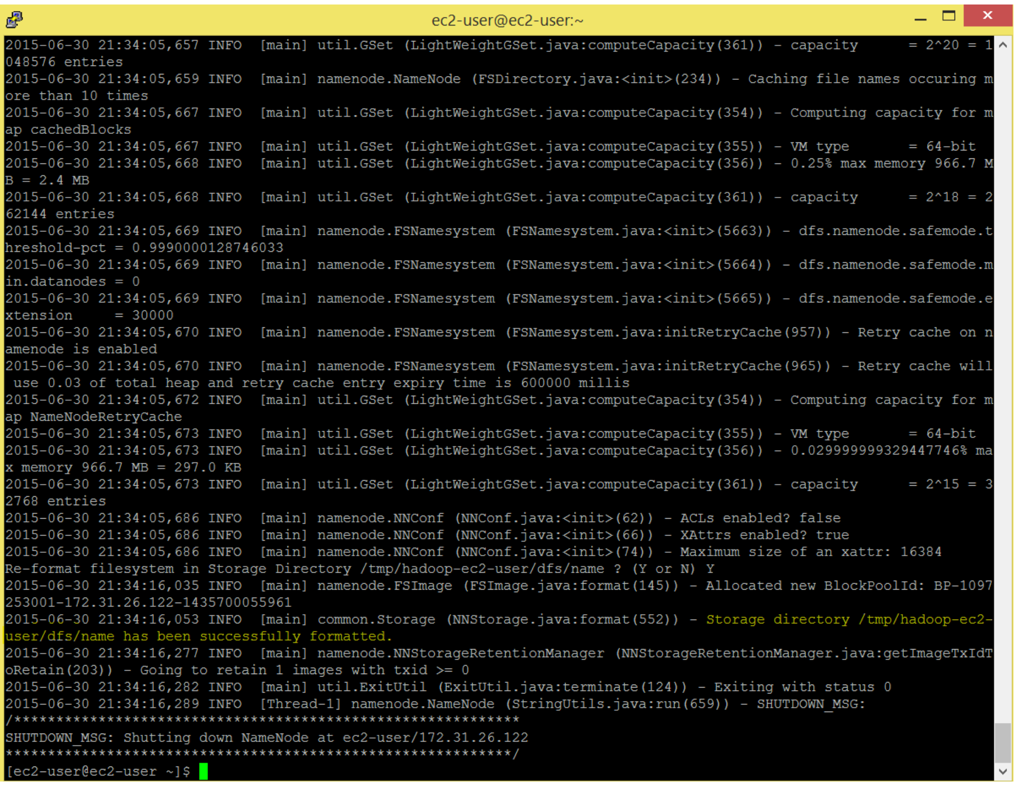
Now we are ready to start the cluster. We will format the namenode.

Command: cd

Command: hadoop namenode -format



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/311.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/311.png)



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/321.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/321.png)

After we format the namenode successfully, we will start all the hadoop daemons.

You are now all set to start the HDFS services i.e. Name Node, Secondary Name Node, and Data Node on your Hadoop Cluster.

Command: cd hadoop-2.6.0/sbin/

Command: ./start-dfs.sh

Start the YARN services i.e. ResourceManager and NodeManager

Command: ./start-yarn.sh

Now run jps command to check if the daemons are running.

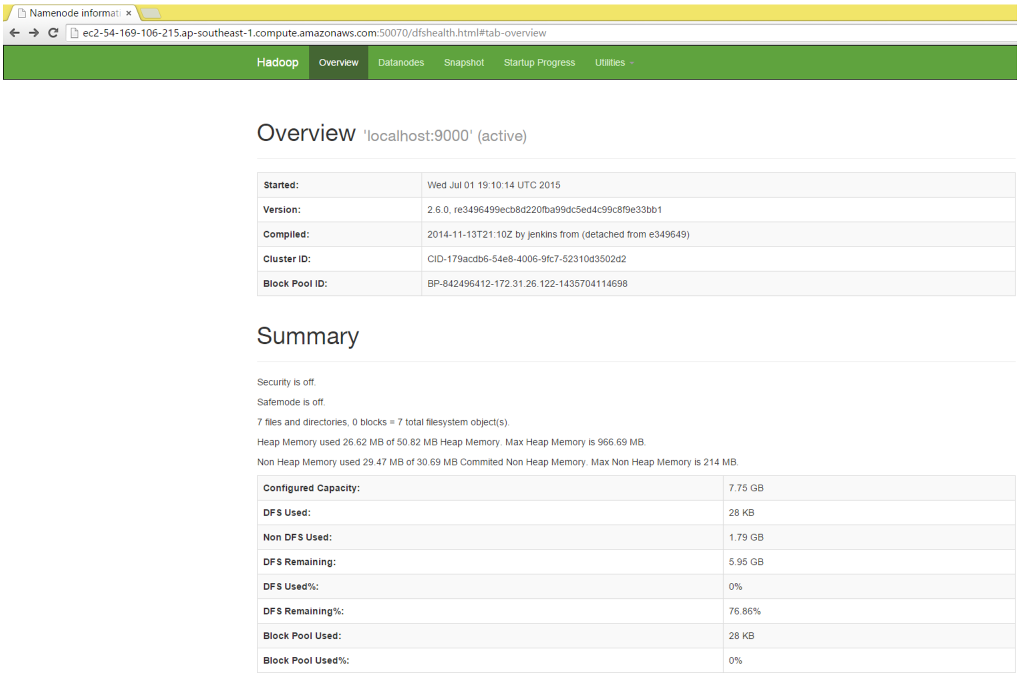
Command: jps

[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/331.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/331.png)

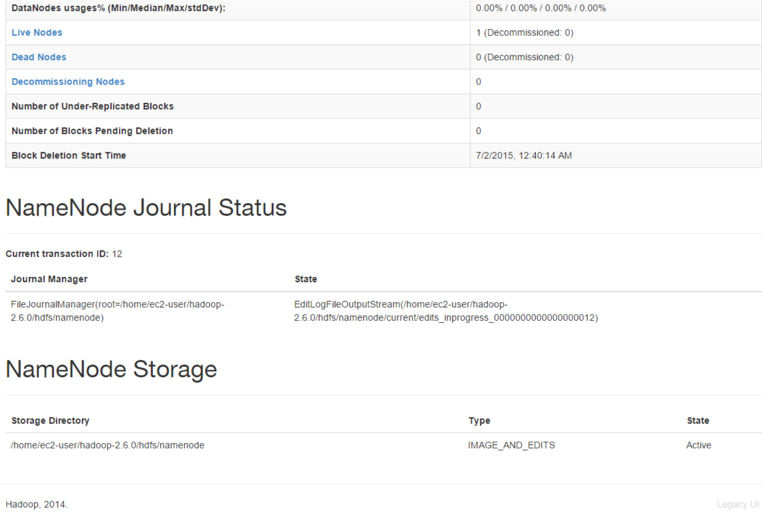
Now open a browser o your system and browse:

ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com:50070

Congratulations your cluster is up and running.



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/341.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/341.png)



[(http://www.eduonix.com/blog/wp-content/uploads/2015/07/35.png)](http://www.eduonix.com/blog/wp-content/uploads/2015/07/35.png)

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(http://www... and-hadoop%2... step-by-step-guide-to-install-hadoop-cluster-on-amazon-ec2%2F&ti... step by step guide to install

Hadoop cluster on Amazon EC2&summ is a step by step guide to install a Hadoop cluster on Amazon EC2. I have my AWS EC2 instance ec2-54-169-106-215.ap-southeast-1.compute ready on which I will install and configure Hadoop, java 1.7 is already installed. In case java is not installed on you AWS EC2 instance,)

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Myles Gartland · Professor at Rockhurst University

in your .bashrc should your "export HADOOP_CONF_DIR=$HOM
dash in it?

"export HADOOP_CONF_DIR=$HOME/hadoop-2.6.0/etc/hadoop

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Naveen Mandadhi · Northern Illinois University

yes, it should! I struggled with it for so long until I figured

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Vijju

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Hello, I followed the steps above. But, am getting below error.

[ec2-user@ip-10-177-1-69 sbin]$ ./start-dfs.sh
Incorrect configuration: namenode address dfs.namenode.servicerpc-address or dfs.namenode.rpc-address is not configured.
Starting namenodes on []
Error: Cannot find configuration directory: /home/ec2-user/hadoop2.6.0/etc/hadoop
Error: Cannot find configuration directory: /home/ec2-user/hadoop2.6.0/etc/hadoop
Starting secondary namenodes [0.0.0.0]
Error: Cannot find configuration directory: /home/ec2-user/hadoop2.6.0/etc/hadoop

Naveen

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I am getting the same error. Please help if you found solution.

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